

Algebra: Properties of Logarithms

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Name: _____ Date: _____

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DIRECTIONS

Evaluate or simplify using Product, Quotient, and Power Rules of logarithms.

1 Evaluate:

$$\log_{10} 1000$$

Answer: _____

2 Apply log properties to simplify:

$$b^{\log_b 5}$$

Answer: _____

3 Apply log properties to simplify:

$$\log_b b^7$$

Answer: _____

4 Apply log properties to simplify:

$$\log_b 1$$

Answer: _____

5 Evaluate:

$$\log_2 1$$

Answer: _____

6 Evaluate:

$$\log_5 125$$

Answer: _____

7 Evaluate:

$$\log_7 7$$

Answer: _____

8 Apply log properties to simplify:

$$\log_{10} 10^x$$

Answer: _____

9 Evaluate:

$$\log_2 32$$

Answer: _____

10 Evaluate:

$$\ln e^5$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Key values: $\log_b(1)=0$, $\log_b(b)=1$, $\log_b(b^n)=n$. These appear frequently on exams.

1 Evaluate:

$$= 3$$
$$\log_{10} 1000$$

2 Apply log properties to simplify:

$$= 5$$
$$b^{\log_b 5}$$

3 Apply log properties to simplify:

$$= 7$$
$$\log_b b^7$$

4 Apply log properties to simplify:

$$= 0$$
$$\log_b 1$$

5 Evaluate:

$$= 0$$
$$\log_2 1$$

6 Evaluate:

$$= 3$$
$$\log_5 125$$

7 Evaluate:

$$= 1$$
$$\log_7 7$$

8 Apply log properties to simplify:

$$= x$$
$$\log_{10} 10^x$$

9 Evaluate:

$$= 5$$
$$\log_2 32$$

10 Evaluate:

$$= 5$$
$$\ln e^5$$



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Page 2